



Edoardo Fiorini

Robotics Engineer

Born: November 13, 1998, Verona, Italy

Address: Via Sinnai, 195, Bovolone (VR), 37051, Italy

Phone: +39 3925439226

E-mail: edoardo.fiorini01@gmail.com

Github: edofiorini

BRIEF BIOGRAPHY

Edoardo Fiorini is a robotics engineer graduated from the University of Verona, Italy, in March 2023. He has a strong background in Programming, Robotics, Control Theory, Computer Vision and Machine Learning. During his training he did several projects where the overall aim is to investigate how those fields can be merged together for building an intelligent system. Moreover, he spent a period abroad for an Internship at Deutsche Zentrum für Luft- und Raumfahrt (DLR), in Monaco, within the Factory of the Future (FoF) development team where he mainly worked for a demo of an intelligent production line showed at Automatica trade fair. Up to now his areas of interest are Industrial Robotics, Control and Systems Theory, Anomaly Detection, Learning by Demonstration and Teleoperation. He is enthusiastic and curious to extend his skills in other fields such as surgical and agricultural robotics.

PROFESSIONAL EXPERIENCE

May 2023 – Present

Robotics researcher at ALTAIR Robotics Lab, University of Verona

- Developing of an autonomous rover with a manipulator mounted on top for precision agriculture.
- Several topics are investigated such as fusion sensor (Lidar, GPS, camera and radar) for navigation, motion planning, obstacle avoidance and fruit recognition.
- The overall system is based on ROS2 environment, which consists of several nodes developed in C++, Python, and MatLab languages.

August 2022 – February 2023

Master's Thesis at ALTAIR Robotics Lab, University of Verona

- Implementation and validation of a robust motion planning workflow for a welding robot based on Learning by Demonstration paradigm through a hand-tool pointer detected by an optical system and the welding workpiece CAD model.
- The experimental collaborative welding cell integrates the robotic manipulator Franka Emika Panda and the Optitrack motion capture system.
- The overall system is based on ROS environment, which consists of several nodes developed in C++, Python, and MatLab languages.

October 2022 – January 2023

Graduate Teaching Assistant

- Teaching Assistant for the class of Programming I, Bachelor's degrees in Computer Science, University of Verona, Italy
- C Language
- Reviewed theoretical concepts, proposed exercises and code, held office hours and invigilated exams.

April 2022 – June 2022

Internship at Deutsche Zentrum für Luft- und Raumfahrt(DLR), Institut für Robotik und Mechatronik, Oberpfaffenhofen, Munich

- Participation in the development team Factory of the Future (FoF) for a new demonstration of a flexible production network showed at Automatica trade fair
- Investigation of motion planning methods for the robotic manipulator LWR mounted on a linear axis
- Implementation and integration of motion planning into the robotic system and experimental evaluation of the method
- Python Language

October 2021 – January 2022

Graduate Teaching Assistant

- Teaching Assistant for the class of Programming Languages and Programming tools with laboratory, Bachelor's degrees in Human Centered Medical System Engineering, University of Verona, Italy
- C Language
- Reviewed theoretical concepts, proposed exercises and code, held office hours and invigilated exams.

March 2021 – June 2021

Graduate Teaching Assistant

- Teaching Assistant for the class of Programming I, Bachelor's degrees in Computer Science, University of Verona, Italy
- C Language
- Reviewed theoretical concepts, proposed exercises and code, held office hours and invigilated exams.

July 2019 – February 2020

Student Collaboration

- Assistant in the Library "Luigi Ambrosoli", Culture and Civilization Department, University of Verona, Italy
- Management of books and related information.

EDUCATION

2020 – 2022

Master's Degree in Computer Engineering for Robotics and Smart Industry, University of Verona, Italy

- Degree Class: Computer Engineering LM-32
- Thesis: A robust motion planning algorithm for Welding Robots based on Optical Tracked Learning by Demonstration and CAD model
- Supervisor: Prof. Riccardo Muradore
- Score: 110/110 cum laude

2017 – 2020

Bachelor's Degree in Bioinformatics, University of Verona, Italy

- Degree Class: Computer Science L-31
- Thesis: Robotics Interface for a BCI System
- Supervisor: Dr. Francesco Setti
- Score: 107/110

2012 – 2017

High School

- Liceo scientifico Statale "Galileo Galilei", Verona, Italy
- Score: 79/100

CERTIFICATIONS

11 December 2020

English Language Certification

- TOEIC, Level B2

May 2016 – June 2016

Safety Certification

- Safety General training for workers

TECHNICAL SKILLS

Programming Language

- MatLab
- Python
- Java
- C/C++
- HTML
- Latex
- SQL, Database

Operating System

- Linux
- Windows
- MacOS

Technologies worked with

- Robot Operating System (ROS)
- Manipulator UR5, Universal Robot
- Manipulator Kuka iiwa, Kuka
- Manipulator Panda, Franka Emika
- Mobile Robot RB-Kairos, Robotnik
- Scanner 3D Gocator
- Simulink

PROJECTS

August 2022 – February 2023

A robust motion planning algorithm for Welding Robots based on Optical Tracked Learning by Demonstration and CAD model

Master's thesis in collaboration with the ALTAIR Robotics Laboratory, under the supervision of Prof. Riccardo Muradore. Investigation, implementation and validation of a novel motion planning algorithm for collaborative welding cell allowing the human-operator to teach to the robot the desired trajectory using an hand-tool pointer which motion is tracked by an optical system. The overall aim of the thesis was to improve in terms of handling the already existing welding cell on the market. All the code is based on ROS/Python/C++ and the experimental setup is composed of Franka Emika Panda and Optitrack motion capture system.

August 2022 – September 2022

Mobile Robotics

"Mobile Robotics" class' final project. Investigate the anomaly detection technique based on PCA reconstruction and Mahalanobis distance decomposition on mobile robot kairos. The overall aim is not just to rise an alert at certain time instant, but to also understand which is the sensor responsible for the fault. The system is able to take the data as input and to give as result the meaning of the anomaly. All the code has been developed in ROS/Python.

January 2022 – February 2022

Process Monitoring

"Process Monitoring" class' final project. Build of the Industrial Robot Anomaly Detection (IndRAD) dataset as a benchmark for evaluating fault detection algorithms on industrial robots. The dataset is composed by 13 nominal trajectories and 3 trajectories with structural anomalies acquired on KUKA LBR iiwa robot. Implementation and investigation of a protocol to inject sensory anomalies in clean data. Explore three anomaly detection methods reported in the literature namely PCA based, deep autoencoders, and one-class classification. The overall work has been presented on ICPR workshops 2022. All the code has been developed in Python.

November 2021 – January 2022

Physical Human-Robot Interaction

"Physical Human-Robot Interaction" class' final project. Investigate and design of different teleoperation approaches such as Four-channel, Scattering-based and Tank-based bilateral teleoperation architectures, taking in to account the passivity theoretical concept. Develop of Kalman filter and Linear methods for regression both applied on real motor data. All the code has been implemented in MatLab/Simulink.

November 2021 – January 2022

Advanced Control Systems

"Advanced Control Systems" class' final project. Compute the Lagrangian and Newton-Euler dynamic model of a 3-DoF robot. Design of Gravity compensation and Inverse Dynamic motion controllers both in joint space and Operational Space. Design of Compliance, Impedance, Admittance, direct force control with inner position loop and parallel position/force controllers. All the code has been implemented in MatLab/Simulink.

September 2021 – October 2021

Robotics, Vision and Control

"Robotics, Vision and Control" class' final project. Design of point to point trajectories and multi-point trajectories in joint space (polynomial, cycloid, harmonic, trapezoidal, double S). Design of a trajectory in operational space using primitive as linear and circular motion. Moreover, there is a part dedicated to the 3D analysis, image analysis and hand-eye calibration. All the set of different motion planning trajectory has been tested for a UR5 model developed in MatLab/Simulink.

September 2021

Gaussian Mixture Model-Hidden Markov Model(GMM-HMM) vs Linear Regression(LR) for Cryptocurrencies Forecasting with new features extraction

"Machine Learning and Artificial Intelligence" class' final project. Forecast of the next day's closing price of a cryptocurrency. The results highlight that the model based on GMM-HMM predicts better because it takes into account all the temporal series before. All the code has been implemented in Python using the related available libraries.

July 2021 – September 2021

High level control of UR5e with ROS and MoveIt both in simulation and reality

"Robot, Programming and Control" class' final project. Program of a flexible industrial cell based on ROS. The key points addressed are: geometric modeling of a UR5e in a manufacturing cell, object Pick and place, glue sealing (following a cartesian path with trajectory constraints in orientation and speed), quality inspection (pick an object and show it with two different angles to a camera), communication with the simulator (URsim) and the real robot. The trajectory are planned with MoveIt. Everything has been implemented in C++. The project was in collaboration with ICE (Industrial Computer Engineering) laboratory, University of Verona.

March 2020 – June 2020

Robotics Interface for a BCI System

Bachelor's Degrees Thesis, collaboration with "Vision, Image Processing Sound (VIPS)" laboratory, University of Verona, under the supervision of Dr. Francesco Setti and Prof. Silvia Francesca Storti. Design the interface between signals from EMG acquisition and UR5 robot. Based on ROS/Python and V-Rep as simulator.

PUBLICATIONS

- Fiorini et al., IndRAD: A Benchmark for Anomaly Detection on Industrial Robots, ICPR workshops 2022

PERSONAL SKILLS

- Team Worker
- Leadership
- Enthusiast
- Manager
- Flexible

Languages

- Italian: Native speaker
- English : Fluent, B2 certification

Others

- Driver license B

OTHER ACTIVITIES

- More than 10 years of experience in the management of parish recreational activities
- Amateur football player
- Passionate about photography and live music concerts
- Travel enthusiast

I authorise the handling of my personal data pursuant to the Personal Data Protection Code – Legislative Decree n. 196/2003.

Verona, 08/08/2023

Edoardo Fiorini